Work, Energy and Power — Worked Solutions

Numericals Worksheet, Grade IX (CBSE) · NCERT Science Chapter: Work and Energy

Key ideas you need (NCERT)

1. **Work** is done when a force acts on an object and the object is **displaced**. For a constant force acting along the direction of displacement: **W = F × s**.
2. **SI unit of work** is the **joule (J)**: 1 J = 1 N × 1 m. Always convert distances to metres (cm ÷ 100, km × 1000).
3. Work is **positive** when force and displacement are in the same direction, **negative** when they are opposite (e.g. friction), and **zero** when there is no displacement or the force is perpendicular to the displacement.
4. **Work done against gravity** in lifting a mass m through a height h: **W = m g h** (this equals the potential energy gained).
5. **Force at an angle θ** to the displacement: only the component F cos θ does work, so **W = F s cos θ**. (Note: the NCERT Class IX text uses only W = F s along the force and the 0°/90°/180° cases. Questions 11–13 and 16–18 use cos θ, which is an extension beyond the Class IX syllabus — useful practice for Class XI.)

Angle θ	0°	30°	45°	60°	90°	180°
cos θ	1	0.866	0.707	0.5	0	−1

Solutions

Q1. A force of 25 N moves a box through a distance of 4 m in the direction of the force. Calculate the work done.

Given: F = 25 N, s = 4 m (force and displacement in the same direction).

W = F × s = 25 N × 4 m

Answer: W = 100 J

Q2. A boy pushes a box with a force of 60 N and the box moves 5 m in the direction of the push. Find the work done by the boy.

Given: F = 60 N, s = 5 m.

W = F × s = 60 × 5

Answer: W = 300 J

Q3. A force of 20 N does 500 J of work on a trolley. Through what distance did the trolley move?

Given: W = 500 J, F = 20 N. Find s.

W = F × s ⇒ s = W ÷ F = 500 ÷ 20

Answer: s = 25 m

Q4. A trolley is displaced by 12 m when a constant force does 360 J of work on it. Find the magnitude of the force.

Given: W = 360 J, s = 12 m. Find F.

F = W ÷ s = 360 ÷ 12

Answer: F = 30 N

Q5. A student lifts a 5 kg book from the floor to a shelf 2 m high. Work done against gravity? (g = 10 m/s²)

Given: m = 5 kg, h = 2 m, g = 10 m/s².

Force needed = weight = m g = 5 × 10 = 50 N (acting upward, displacement upward).

W = m g h = 50 N × 2 m

Answer: W = 100 J

Q6. A coolie lifts a 20 kg load from the ground and places it on his head at a height of 1.5 m. Work done? (g = 10 m/s²)

Given: m = 20 kg, h = 1.5 m, g = 10 m/s².

W = m g h = 20 × 10 × 1.5

Answer: W = 300 J

Note: This work is stored in the load as gravitational potential energy (Ep = mgh).

Q7. A horse pulls a cart with a constant force of 500 N over a distance of 2 km. Work done in joules?

Given: F = 500 N, s = 2 km = 2 × 1000 = 2000 m (convert to SI first!).

W = F × s = 500 × 2000

Answer: W = 1,000,000 J = 10^6 J (= 1 MJ)

Q8. A man pushes against a rigid wall with a force of 200 N but the wall does not move. Work done on the wall?

Given: F = 200 N, s = 0 (wall does not move).

W = F × s = 200 × 0

Answer: W = 0 J

Note: Although the man gets tired (his muscles use energy), in the scientific sense no work is done on the wall because there is no displacement.

Q9. A girl carries a 12 kg bucket of water up a staircase to a height of 8 m. Work done against gravity? (g = 10 m/s²)

Given: m = 12 kg, h = 8 m.

W = m g h = 12 × 10 × 8

Answer: W = 960 J

Note: Only the vertical height matters for work against gravity, not the length or shape of the staircase.

Q10. A force of 8 N acts on a body and moves it 250 cm in the direction of the force. Work done in joules?

Given: F = 8 N, s = 250 cm = 250 ÷ 100 = 2.5 m.

W = F × s = 8 × 2.5

Answer: W = 20 J

Q11. A force of 40 N acts at 60° to the horizontal and moves a body 10 m horizontally. Calculate the work done.

Given: F = 40 N, s = 10 m, θ = 60°, cos 60° = 0.5.

Effective force along displacement = F cos θ = 40 × 0.5 = 20 N.

W = F s cos θ = 40 × 10 × 0.5

Answer: W = 200 J

Q12. A force of 50 N inclined at 30° to the horizontal displaces a block by 8 m along the horizontal. (cos 30° = 0.866)

Given: F = 50 N, s = 8 m, θ = 30°.

W = F s cos θ = 50 × 8 × 0.866 = 400 × 0.866

Answer: W = 346.4 J

Q13. A person pushes a lawn mower with a force of 100 N directed at 45° to the ground. Work done in moving it 20 m? (cos 45° = 0.707)

Given: F = 100 N, s = 20 m, θ = 45°.

W = F s cos θ = 100 × 20 × 0.707 = 2000 × 0.707

Answer: W = 1414 J

Q14. A porter carries a 30 N bag and walks 5 m on a level road. The force on the bag is vertical (perpendicular to displacement). Work done by the porter on the bag?

Force (upward, 30 N) and displacement (horizontal, 5 m) are at θ = 90°; cos 90° = 0.

W = F s cos 90° = 30 × 5 × 0

Answer: W = 0 J

Note: NCERT case: when force is perpendicular to the direction of motion, no work is done by that force.

Q15. A frictional force of 20 N acts opposite to the motion of a block that slides 5 m. Work done by friction?

Friction is opposite to displacement, so θ = 180° and cos 180° = −1.

W = F s cos 180° = 20 × 5 × (−1)

Answer: W = −100 J

Note: Negative work means the force opposes the motion and removes energy from the block (as heat).

Q16. A child pulls a sled with a rope inclined at 60° to the ground with a force of 25 N. Work done in pulling the sled 12 m?

Given: F = 25 N, s = 12 m, θ = 60°, cos 60° = 0.5.

W = F s cos θ = 25 × 12 × 0.5

Answer: W = 150 J

Q17. A force of 60 N does 150 J of work when the displacement is 5 m. Find the angle between the force and the displacement.

Given: W = 150 J, F = 60 N, s = 5 m.

W = F s cos θ ⇒ cos θ = W ÷ (F s) = 150 ÷ (60 × 5) = 150 ÷ 300 = 0.5

cos θ = 0.5 ⇒ θ = 60°

Answer: θ = 60°

Q18. A force of 20 N moves an object 10 m and does 173.2 J of work. Find the angle between force and displacement. (cos 30° = 0.866)

Given: W = 173.2 J, F = 20 N, s = 10 m.

cos θ = W ÷ (F s) = 173.2 ÷ (20 × 10) = 173.2 ÷ 200 = 0.866

cos θ = 0.866 ⇒ θ = 30°

Answer: θ = 30°

Answer summary

Q	Answer	Q	Answer	Q	Answer
1	100 J	7	10^6 J	13	1414 J
2	300 J	8	0 J	14	0 J
3	25 m	9	960 J	15	−100 J
4	30 N	10	20 J	16	150 J
5	100 J	11	200 J	17	60°
6	300 J	12	346.4 J	18	30°

Exam tips (CBSE)

• Always write: Given → Formula → Substitution (with units) → Answer with SI unit. Marks are given for each step.
• Convert to SI units first: cm → m, km → m, g → kg.
• Remember the special cases: no displacement ⇒ W = 0; force ⊥ motion ⇒ W = 0; force opposite motion ⇒ W negative.
• Work done against gravity depends only on vertical height (W = mgh).
• All 18 answers above agree with the answers printed on the worksheet.